

$$\frac{9}{9} \Rightarrow 100\%$$

0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
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Name: _____

These questions assess your understanding of the material from Chapters 23, 24 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. How much energy is stored in a 4.5-mH inductor when a 16.-A current flows through it?

$$E = \frac{1}{2} L I^2$$

ANSWER: 0.576 J ✓

$$E = \frac{1}{2} (4.5 \times 10^{-3}) (16)^2$$

$$= 0.576 \text{ J}$$

2. On its highest power setting, a 900-W microwave oven projects microwaves onto a 0.27-m-by-0.27-m area. Calculate the peak electric-field strength in the electromagnetic waves emitted by the microwave oven.

$$I = \frac{C \epsilon_0 E_0^2}{2} = \frac{P}{A}$$

ANSWER: 3049.58 $\frac{\text{V}}{\text{m}}$ ✓

$$\Rightarrow E_0 = \sqrt{\frac{2P}{C \epsilon_0 A}} = \sqrt{\frac{2(900)}{(3 \times 10^8)(8.85 \times 10^{-12})(0.27)^2}} = \sqrt{\frac{1800}{1.935495 \times 10^{-4}}}$$

$$E_0 = 3049.58 \frac{\text{V}}{\text{m}}$$

3. A portable x-ray unit has a step-up transformer, the 110V-input of which is transformed to the 2.0×10^5 -V output needed by the x-ray tube. The primary has 74. loops and draws a current of 2.8 A when in use. Find the current output of the secondary.

$$\frac{V_s}{V_p} = \frac{I_p}{I_s} \Rightarrow I_s = I_p \frac{V_p}{V_s}$$

ANSWER: $1.54 \times 10^{-3} \text{ A}$ ✓

$$\Rightarrow I_s = (2.8) \left(\frac{110}{2 \times 10^5} \right) = 1.54 \times 10^{-3} \text{ A}$$

4. What is the characteristic time constant for a 9.7-mH inductor in series with a 1.7- Ω resistor?

$$\tau = \frac{L}{R} = \frac{9.7 \times 10^{-3}}{1.7} = 5.706 \times 10^{-3} \text{ s}$$

ANSWER: $5.706 \times 10^{-3} \text{ s}$ ✓

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5. A generator coil is rotated 0.91 revolutions in 31. ms. The 180-turn circular coil has a 4.7-cm radius and is in a uniform 2.5-T magnetic field. What is the average emf that is induced?

$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t} \Rightarrow \Delta \Phi = BA(\cos \theta_f - \cos \theta_o)$$

ANSWER: 15.68 V

$$\Rightarrow \mathcal{E} = -N \cdot \frac{BA(\cos 327.6 - \cos 0)}{\Delta t} = -180 \cdot \frac{(2.5)(\pi (0.047)^2)(-0.1557)}{31 \times 10^{-3}}$$

$$\mathcal{E} = 15.68 \text{ V}$$

6. What is the maximum strength of the magnetic field in an electromagnetic wave that has a maximum electric-field strength of 530 V/m?

$$c = \frac{E}{B} \Rightarrow B = \frac{E}{c} = \frac{(530)}{3 \times 10^8}$$

ANSWER: $1.767 \times 10^{-6} \text{ T}$

$$B = 1.767 \times 10^{-6} \text{ T}$$

7. A battery charger meant for a series connection of several nickel-cadmium batteries needs to have a 12.-V output to charge the batteries. It uses a step-down transformer with a 170-loop primary and a 120-V input. If the charging current is 13. A, what is the input current?

$$\frac{V_s}{V_p} = \frac{I_p}{I_s} \Rightarrow I_p = I_s \times \frac{V_s}{V_p}$$

ANSWER: 1.3 A

$$I_p = (13) \left(\frac{12}{120} \right) = 1.3 \text{ A}$$

8. Suppose a 0.25-H inductor is in series with a 1.2- Ω resistor and a current of 3.5 A flows through them. Find the current that flows through the inductor and resistor 4.5 s after the battery is disconnected.

$$I(t) = I_0 e^{-t/\tau} \quad \tau = \frac{L}{R}$$

$$\Rightarrow I(t) = I_0 e^{-\left(\frac{tR}{L}\right)}$$

$$I(t) = (3.5) e^{-\left(\frac{4.5 \times 1.2}{0.25}\right)} = 1.456 \times 10^{-9} \text{ A}$$

ANSWER: $1.456 \times 10^{-9} \text{ A}$

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9. Calculate the self-inductance of a 0.14 m-long, 0.052 m-diameter solenoid that has 430 coils.

$$L = \frac{\mu_0 N^2 A}{l} \quad r = 0.026$$

ANSWER: $\underline{3.5246 \times 10^{-3} \text{ H}}$

$$L = \frac{(4\pi \times 10^{-7})(430^2)(\pi \times 0.026^2)}{0.14}$$

$$= 3.5246 \times 10^{-3} \text{ H}$$

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$$1. E_{ind} = 0.576 \text{ J} \therefore E_{ind} = \frac{1}{2} L I^2$$

$$2. E_0 = 3.05 \times 10^3 \frac{\text{V}}{\text{m}} \therefore I_{ave} = \frac{c \epsilon_0 E_0^2}{2} = \frac{c B_0^2}{2 \mu_0} = \frac{E_0 B_0}{2 \mu_0}$$

$$3. I_s = 1.54 \times 10^{-3} \text{ A} \therefore \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

$$4. \tau = 5.71 \times 10^{-3} \text{ s} \therefore \tau = \frac{L}{R}$$

$$5. \mathcal{E} = 15.7 \text{ V} \therefore \mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

$$6. B_{max} = 1.77 \times 10^{-6} \text{ T} \therefore c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{E}{B} = f \lambda$$

$$7. I_p = 1.30 \text{ A} \therefore \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

$$8. I = 1.46 \times 10^{-9} \text{ A} \therefore I(t) = I_0 e^{-t/\tau}, \tau = \frac{L}{R}$$

$$9. L = 3.52 \times 10^{-3} \text{ H} \therefore L = N \frac{\Delta \Phi}{\Delta I} = \frac{\mu_0 N^2 A}{\ell}$$